

Chart and Section Summaries

Part 1:

Top 10 countries AI adoption:

The top 10 ranking shows that enterprise AI adoption is led primarily by Nordic and Western European countries, with Denmark standing out as the strongest adopter. The leading group includes countries with well-developed digital economies, while the presence of Estonia also shows that smaller, digitally advanced countries can achieve high business AI uptake.

For the wider project, these countries represent the areas where AI-related demand and supporting digital infrastructure may grow most quickly. However, high adoption alone does not determine sustainability: the environmental implications depend on whether each country's energy system and data-centre infrastructure are supported by a strong renewable-energy base.

AI Adoption percentage:

Average enterprise AI adoption increased substantially across the countries in the dataset, with relatively limited growth between 2021 and 2023 followed by a much sharper rise in 2024 and 2025. This suggests that business use of AI is moving from early adoption toward more widespread implementation across Europe.

The chart supports the project's central question because a faster expansion of enterprise AI could increase demand for digital infrastructure and electricity. However, the line should be interpreted carefully: 2022 is not available in the dataset, so it shows reported observation years rather than a fully continuous annual trend.

Part 3B:

Datacenters vs Renewable energy:

The chart shows no clear overall relationship between the number of listed data-centre facilities and renewable-energy share. Several countries with the largest data-centre presence have relatively low to moderate renewable shares, while some of the strongest renewable-energy systems have fewer listed facilities. This suggests that data-centre infrastructure is concentrated in countries with different energy profiles, and a larger facility footprint does not automatically mean it is supported by a cleaner energy base. Because the measure is a count of listed facilities rather than capacity or electricity use, it indicates infrastructure presence rather than direct environmental impact.

3B Summary: The 3B analysis shows that a larger number of listed data-centre facilities does not automatically correspond to a stronger renewable-energy base. Data-centre infrastructure is spread across countries with very different energy profiles: some appear relatively well positioned to support digital growth with higher renewable shares, while others combine substantial infrastructure with less renewable-backed energy systems. This suggests that the sustainability of expanding digital infrastructure depends not only on where facilities are concentrated, but also on how quickly national energy systems transition toward renewables. Because the data measures listed facilities rather than capacity or electricity use, the findings describe infrastructure presence rather than direct environmental impact.

Part 4B

Datacenters vs CO₂:

The chart shows a broad positive relationship between listed data-centre facilities and power-sector CO₂ emissions: countries with larger data-centre footprints tend to appear among the higher-emitting countries. However, this pattern is strongly influenced by large national economies and electricity systems, rather than showing that data centres themselves are the direct cause of emissions.

Several countries with relatively few listed facilities still have high emissions, while others with substantial infrastructure have more moderate emissions. This suggests that digital

infrastructure is only one part of the picture; country size, industrial activity, electricity demand, and the national energy mix are all likely to shape total power-sector emissions.

4B Summary: The data-centre comparison does not show a simple one-to-one relationship between the number of listed facilities and power-sector CO₂ emissions. Countries with a larger data-centre presence can have very different emissions profiles, suggesting that national emissions are shaped more strongly by factors such as country size, electricity demand, industry, and energy mix. Listed facility count is therefore a useful measure of infrastructure presence, but it does not measure data-centre capacity, electricity use, or AI-specific environmental impact.

5B:

Datacenters by AI category:

This chart adds a data-centre infrastructure lens to the original AI sustainability categories. Sustainable AI Leaders generally combine strong renewable-energy shares with modest-to-medium listed data-centre footprints, suggesting that favourable energy conditions can coexist with active AI adoption without necessarily requiring the largest facility concentration. In contrast, Environmental Risk countries remain below the renewable-energy benchmark and include some of the largest listed data-centre footprints, reinforcing the concern that infrastructure growth may create greater pressure where the energy system is less renewable-backed.

Green Potential countries tend to have stronger renewable conditions but smaller current data-centre footprints, which may position them for cleaner future digital expansion. Developing Transition countries generally combine lower renewable shares with more limited infrastructure, indicating that both digital development and energy transition may still need to progress together. Importantly, the colours come from the original AI-adoption and renewable-energy categories—not from data-centre counts—so the chart shows how current infrastructure is distributed across those existing sustainability profiles.

AI category heatmap:

The table highlights a clear sustainability divide between the categories. Environmental Risk countries have the largest average listed data-centre presence while also showing the lowest renewable-energy share and highest average CO₂ emissions, making this group the clearest potential pressure point as digital infrastructure expands. In contrast, Sustainable AI Leaders combine a meaningful data-centre presence with the strongest renewable-

energy base and lowest average emissions, suggesting comparatively more favourable conditions for supporting AI and digital growth.

Green Potential countries currently have relatively limited listed infrastructure but a stronger renewable base, which may give them a better foundation for cleaner future expansion. Meanwhile, Developing Transition countries have a larger average data-centre presence than Green Potential but weaker renewable shares and relatively high emissions, indicating that energy-system improvements may need to keep pace with infrastructure growth. These are country-level patterns rather than evidence of direct data-centre emissions, since facility counts do not measure capacity or electricity use.

5B Summary: The 5B analysis reinforces the original AI sustainability story by adding a data-centre infrastructure lens. Countries already classified as Environmental Risk tend to combine the largest listed data-centre presence with weaker renewable-energy shares, suggesting that digital infrastructure growth may place greater pressure on less renewable-backed energy systems. Sustainable AI Leaders show a more balanced position, combining a meaningful infrastructure footprint with stronger renewable-energy conditions. Meanwhile, Green Potential countries appear to have a cleaner foundation for future expansion, while Developing Transition countries may need energy-system improvements to keep pace with their existing infrastructure. These results describe country-level patterns only: listed facilities are not a measure of data-centre capacity, electricity use, or direct emissions.

Part 6B

correlation_matrix_2024: The heatmap shows that current listed data-centre facilities have the strongest relationship with average CO₂ emissions across the countries in this dataset. This suggests that countries with a larger digital-infrastructure footprint also tend to have higher overall emissions, although this likely reflects broader national factors such as country size, electricity demand, industrial activity, and digital-hub status rather than data centres alone.

In contrast, average AI adoption has little relationship with either emissions or data-centre counts. Renewable-energy share has a weak negative relationship with both data-centre facilities and emissions, suggesting that stronger renewable systems may be associated

with somewhat lower environmental pressure, but this pattern is not strong enough to be predictive on its own.

6B Summary: The 6B analysis indicates that the sustainability implications of AI growth appear to be more closely connected to underlying digital infrastructure and national energy conditions than to enterprise AI adoption alone. AI adoption itself does not show a meaningful relationship with emissions, whereas countries with more listed data-centre facilities tend to have substantially higher average CO₂ emissions. At the same time, renewable-energy share shows a modest inverse relationship with emissions, supporting the idea that cleaner energy systems can help reduce the environmental pressure associated with digital growth.

These results show association rather than causation. Listed facilities do not measure data-centre capacity, electricity use, or AI-specific emissions, and the emissions comparison should remain conditional on confirming that the source uses a consistent unit across all countries and years.